Problem 3 — Simulation vs Practical Development

Technical Proposal: Development Approach for the National Aeromodelling Competition RC Aircraft

Prepared by the lead designer

1. Recommendation in brief

I recommend a **hybrid, simulation-informed, prototype-validated** approach. Low-fidelity simulation (XFLR5/OpenVSP plus spreadsheet calculations) is used first, and only briefly, to narrow the design space cheaply. Cheap physical prototypes then follow quickly, because they are the only thing that can tell us whether the simulation is right. Neither pure approach fits a team with limited time, money and resources: pure simulation risks trusting a model that was never checked, and pure prototyping burns scarce balsa, foam, motors and weeks on avoidable mistakes.

Assumptions: four weeks of development; a budget for roughly 2–3 airframes and one spare set of electronics; hand-built foam/balsa/carbon-rod construction; at least one outdoor test field; a team of 4–6 with mixed CAD, electronics and piloting skills; a competition scored on payload and/or maneuverability (see the airfoil problem).

2. Advantages and limitations of each approach

Simulation-led design

- *Advantages:* very cheap per iteration; hundreds of configurations (airfoil, wing area, tail volume, CG) can be compared in hours; no risk of crashing; reveals trends such as the drag polar, stall angle, static margin and trim.
- *Limitations:* accuracy depends on assumptions. At the low Reynolds numbers of small RC aircraft (about 1×10^5 to 3×10^5), laminar separation bubbles, propeller slipstream effects and real build quality are poorly captured by panel/vortex-lattice codes. High-fidelity CFD (ANSYS) is slow, needs skill and licences, and still needs validation. Results can look precise while being wrong (false confidence).

Prototype-led design

- *Advantages:* captures everything real: build tolerances, weight growth, servo slop, propeller-wing interaction, structural flex and pilot feel. Failures are unambiguous.
- *Limitations:* each iteration costs materials and days; crashes destroy parts; weather limits testing; results are hard to attribute (if it flies badly, which of ten variables is the cause?); the team can only explore a few designs, so it may settle on a local optimum.

3. Comparison table

Criterion	Simulation-led	Prototype-led	Proposed hybrid
Cost	Very low (software is free: XFLR5, OpenVSP)	High: repeated airframes, motors,	Low–moderate: 2–3 airframes only

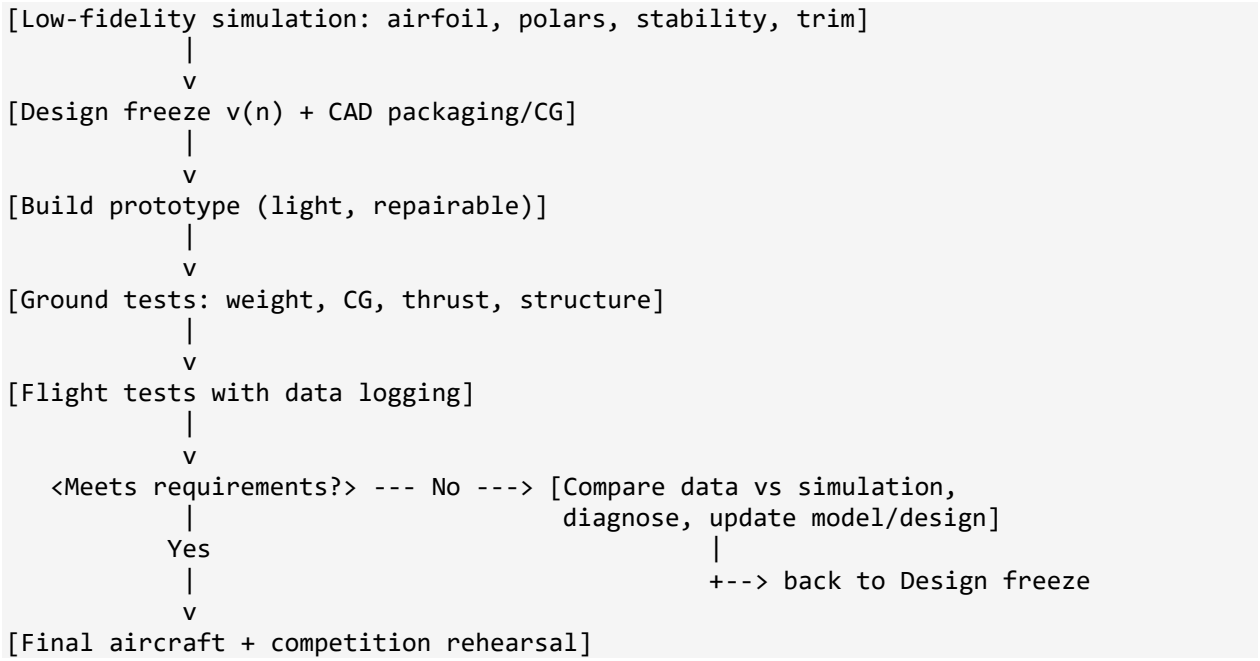
Criterion	Simulation-led	Prototype-led	Proposed hybrid
		crash damage	
Time	Fast per iteration (hours), but setup and learning curve	Slow per iteration (days: build, cure, weather)	Simulation trims the options first, so fewer builds
Accuracy	Good for trends, weak in absolute values at low Re	Highest (it is the real system)	Simulation calibrated against measured data
Iteration speed	Very high	Low	High early, moderate later
Failure detection	Only failures that are modelled; misses structural and integration faults	Detects real faults, but late and sometimes destructively	Aerodynamic and stability faults found early; integration faults found on bench and glide tests
Final performance	Risk of a design that is optimal on paper only	Risk of a locally tuned, unoptimised design	Best expected: optimised region found by simulation, then tuned by flight

4. Proposed workflow

1. **Mission requirements:** take-off distance limit, payload (or maneuverability) target, flight time, wingspan/weight limits from the rules.
2. **Hand calculations:** weight estimate, wing loading, required lift coefficient, stall speed, thrust-to-weight ratio, power and battery sizing, tail volume coefficients.
3. **Simulation (low fidelity):** airfoil comparison, lift/drag polars, neutral point and static margin, trim, rough performance in XFLR5 or OpenVSP. CAD (Fusion 360/SolidWorks) is used for packaging and CG, not for CFD.
4. **Design freeze v1 and build Prototype 1:** simplest, lightest, repairable construction.
5. **Ground tests:** weight and CG check, control-throw and direction check, static thrust and current measurement, structural load test of wing and joints.
6. **Flight tests:** hand-glide, then powered take-off, trim, stall, then payload/maneuver runs with logged data (stopwatch, flight-controller or telemetry log, phone video).
7. **Compare measurements to the simulation, correct the model, refine** and build Prototype 2 (and 3 as the final competition aircraft).
8. **Rehearse** competition tasks with the final aircraft and spare parts.

Flowchart (development loop):

```
[Mission requirements & rules]
            |
            v
[Hand calculations: weight, wing loading, CL, T/W, power]
            |
            v
```



5. Parameters to study in simulation vs verify physically

Studied through simulation (cheap to sweep, trend-driven):

1. **Airfoil and wing geometry:** lift and drag polars, CL max, stall angle, aspect ratio and area trade-offs.
2. **Static stability:** neutral point, static margin, CG range, tail volume coefficients, trim angle.
3. **Performance estimates:** stall speed, lift-to-drag ratio, glide ratio, rough power required vs available.

Verified physically (where models are weakest):

1. **Actual weight and CG:** built aircraft are almost always heavier than modelled, and this changes everything.
2. **Propulsion thrust and current draw:** measured static thrust with the chosen propeller, motor and battery, not catalogue values.
3. **Structural strength and flight behaviour:** wing-root and joint load tests, flutter or flex, take-off run, stall behaviour, control response and handling.

6. Validating the simulation with measurements

- **Weight and CG** measured on a scale, then fed back into the model so that both use the same inputs.
- **Static thrust** from a simple spring/kitchen-scale test; compare with the propulsion model.
- **Glide test:** measure glide ratio (height lost over distance) and compare with the simulated L/D.
- **Stall speed:** from a GPS/telemetry or video-timed pass; compare with the predicted value.
- **Take-off distance and time-to-climb** measured on the field.
- **Acceptance rule:** if measured values are within about 10–15% of predicted, the model is trusted for further trade studies; if not, the model assumptions (drag build-up, Reynolds number,

weight, propeller efficiency) are corrected before the next design iteration. A simulation is trusted only after this calibration.

7. Diagnosis: the simulation flies, the prototype cannot take off

Plausible causes

- Weight is higher than modelled (glue, wiring, battery, paint).
- Thrust is lower than assumed (propeller mismatch, voltage sag, weak battery C-rating, ESC/motor settings).
- CG too far aft or forward, so the aircraft cannot rotate or trim.
- Wrong incidence or thrust angle; warped wing or twist (washout error).
- Airfoil performs worse at real Re and surface roughness than simulated; drag higher than modelled.
- Control-surface problems: reversed or insufficient elevator throw, hinge gaps, servo slop.
- Surface conditions: soft grass or rough runway; wheels too small or too high friction.
- Structural flex changing wing incidence under load.

Systematic diagnostic plan (change one thing at a time)

1. **Inspect and weigh:** compare the real mass and CG with the model; fix the CG first.
2. **Check controls:** direction, throws, neutral position, no binding.
3. **Static thrust test:** measure thrust and current; compare with the model, then fix propulsion if short.
4. **Geometry check:** incidence, twist, symmetry, tail alignment (use a level and a ruler).
5. **Hand-glide test:** if it glides well, the airframe is fine and propulsion/take-off dynamics are the issue; if not, the problem is aerodynamic or CG.
6. **Take-off run test** on a smooth surface, with video to see rotation and stall behaviour.
7. **Update the simulation** with measured inputs (weight, thrust, drag); the revised model should now reproduce the failure, and the fix is then tested in simulation before rebuilding.

8. Bonus: four-week schedule with contingency

Week	Activities	Output
1	Read rules, define requirements, hand calculations, airfoil and configuration simulation, procure materials	Design v1 frozen; BOM ordered
2	Build Prototype 1; ground tests (weight, CG, thrust, structure); first hand-glide and powered flights	Test data, list of issues
3	Compare data with simulation, correct model, redesign; build Prototype 2; repeated flight tests	Validated model; improved aircraft
4	Build/finish final aircraft or upgrade Prototype 2; competition rehearsals; spares packed	Competition-ready aircraft

Contingency (about 20–25% of total time): one buffer day each in weeks 2 and 3 for weather and rebuilds, plus an explicit reserve in week 4 for repairs; one spare wing and tail set, spare propellers and a spare battery are built or bought early. If Prototype 1 fails badly, the team reverts to a simpler, proven configuration instead of redesigning.

9. Conclusion

Simulation is cheapest at the start and prototypes are most informative at the end, so the proposed approach spends effort accordingly. The result is a design narrowed quickly by calculation, then proven and corrected by a small number of real flights, which gives the best chance of a reliable, competitive aircraft within four weeks and a limited budget.